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KLA-Tencor's Leverage Decision:
Cashing in the Chips

"What's it gonna be, Iván? What's the number?" Michelle Shah called across the bullpen in Fletcher Capital Management's temporary offices on Valencia Street in San Francisco. Iván Champeau, battling a headache, winced at the sound of his boss's voice.

Champeau stared at his wall of monitors displaying a Bloomberg terminal, chat windows in which he communicated with other hedge-fund analysts, and the proprietary financial model he had constructed to analyze semiconductor-manufacturing company KLA-Tencor (NASDAQ: KLAC) and its capacity for a leveraged recapitalization transaction. Champeau had been a high achiever at the fund since joining it, in 2010, as his first job out of business school, almost four years ago; but now, in July 2014, he was experiencing a run of bad luck. He hoped to save his career (and year-end bonus) by producing a value-unlocking transaction proposal to KLAC management wherein the almost entirely debtless company would take advantage of historically low interest rates by issuing debt to repurchase its own stock in the open market. Through both organic growth and a string of successful acquisitions, the company had achieved a scale, maturity, and predictability of cash-flow generation that made it an attractive debt issuer to large lenders. The only question that remained was how much debt it could support.

"Come on, Iván—I need a number," Shah said. "How much debt can these guys take on?"

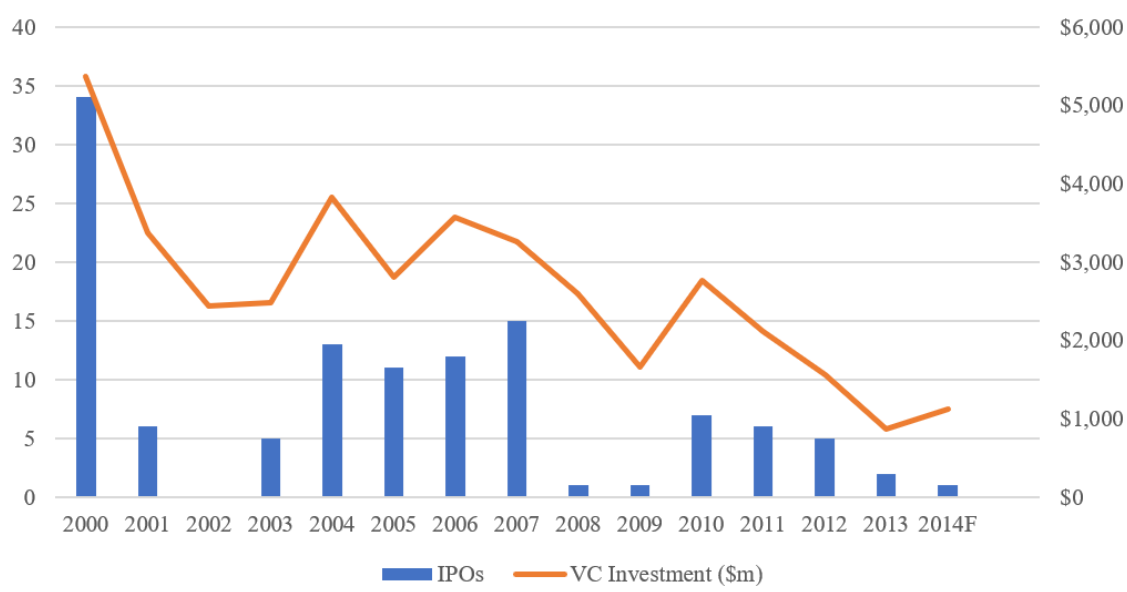
"I'll have it for you in a minute!" Champeau called back, returning to the model.

Semiconductor Capital Equipment Industry Overview

The semiconductor industry comprised companies that designed and manufactured semiconductors and related devices, such as transistors and integrated circuits (or "chips"). These products powered the electronic devices ubiquitous in modern life, from mobile phones to automobiles to "smart home" applications. These chips generally fell into four categories, dominated by logic chips (half of the market), which performed calculations that drove computer processing, and memory chips (one-third), which stored data electronically. Two smaller segments—analog devices and optical sensor devices (OSDs)—comprised about one-sixth of the industry.

Global semiconductor revenues were relatively stable, expected to fall only slightly in 2015, to around $347.3 billion from $354.3 billion a year prior, with IHS Technology forecasting 2.1% total industry revenue growth from 2015 to 2020.[1] The industry historically featured a high degree of competitive rivalry, a moderate-to-high level of supplier concentration, and high barriers to entry due to capital intensity and intellectual property. Large players such as Intel and AMD in the United States, Taiwan's TSMC, and South Korea's Samsung tended to dominate the industry, with other players coming primarily from those countries, plus Japan and the Netherlands. The industry had matured significantly over the past 15 years, partly through aggressive consolidation that afforded significant "moats" to incumbent conglomerates. Because of those high barriers to entry, venture capital investment in the space had fallen by about 80% since the turn of the century, and once-plentiful semiconductor IPOs had become increasingly rare. (See **Figure 1**.)

Figure 1: Semiconductor Venture Capital Investment and IPO Activity, 2000–2014F



Source: Graph created by the case authors based on data from Pitchbook and Thomson Reuters.

1 "Global Semiconductor Market Slumps in 2015, IHS Says," Business Wire, April 4, 2016, https://www.businesswire.com/news/home/20160404005098/en/Global-Semiconductor-Market-Slumps-in-2015-IHS-Says.

KLAC described the semiconductor manufacturing process in its 2014 annual report:

> *The semiconductor fabrication process begins with a bare silicon wafer—a round disk that is typically 150 millimeters, 200 millimeters, or 300 millimeters in diameter, about as thick as a credit card and gray in color. The process of manufacturing wafers is highly sophisticated, involving the creation of large ingots of silicon by pulling them out of a vat of molten silicon. The ingots are then sliced into wafers. Prime silicon wafers are then polished to a mirror finish. Other, more specialized wafers, such as epitaxial silicon ("epi"), silicon on insulator ("SOI"), gallium nitride ("GaN"), and silicon carbide ("SiC") are also common in the semiconductor industry.*
>
> *The manufacturing cycle of an IC* is grouped into three phases: design, fabrication, and testing. IC design involves the architectural layout of the circuit, as well as design verification and reticle generation. The fabrication of a chip is accomplished by depositing a series of film layers that act as conductors, semiconductors, or insulators on bare wafers. The deposition of these film layers is interspersed with numerous other process steps that create circuit patterns, remove portions of the film layers, and perform other functions such as heat treatment, measurement, and inspection. Most advanced chip designs require hundreds of individual steps, many of which are performed multiple times. Most chips consist of two main structures: the lower structure, typically consisting of transistors or capacitors which perform the "smart" functions of the chip; and the upper "interconnect" structure, typically consisting of circuitry which connects the components in the lower structure. When the layers on the wafer have been fabricated, each chip on the wafer is tested for functionality. The wafer is then cut into individual devices, and those chips that passed functional testing are packaged.*[2]

The fabrication process had four primary components:

- Deposition, in which the layers of insulating or conducting materials were deposited onto the silicon wafer
- Lithography, in which lasers or X-rays were shined through a photomask to transfer designs onto the wafer
- Etch-and-clean, in which the deposited material was etched away from the wafer to leave the desired pattern, with the remainder cleaned off to leave a pristine, functional chip
- Process control, wherein each of the first three processes was monitored to remove defects and increase the yield on usable finished chips

The semiconductor capital equipment, or semicap, industry was a sub-industry that provided machinery, testing equipment, and software applications to semiconductor manufacturers to assist in these processes, with companies often focusing on just one or two segments. The semicap industry was highly concentrated for the same reasons that semiconductor manufacturing was, with Applied Materials and Lam Research dominating the deposition segment, ASML and Nikon

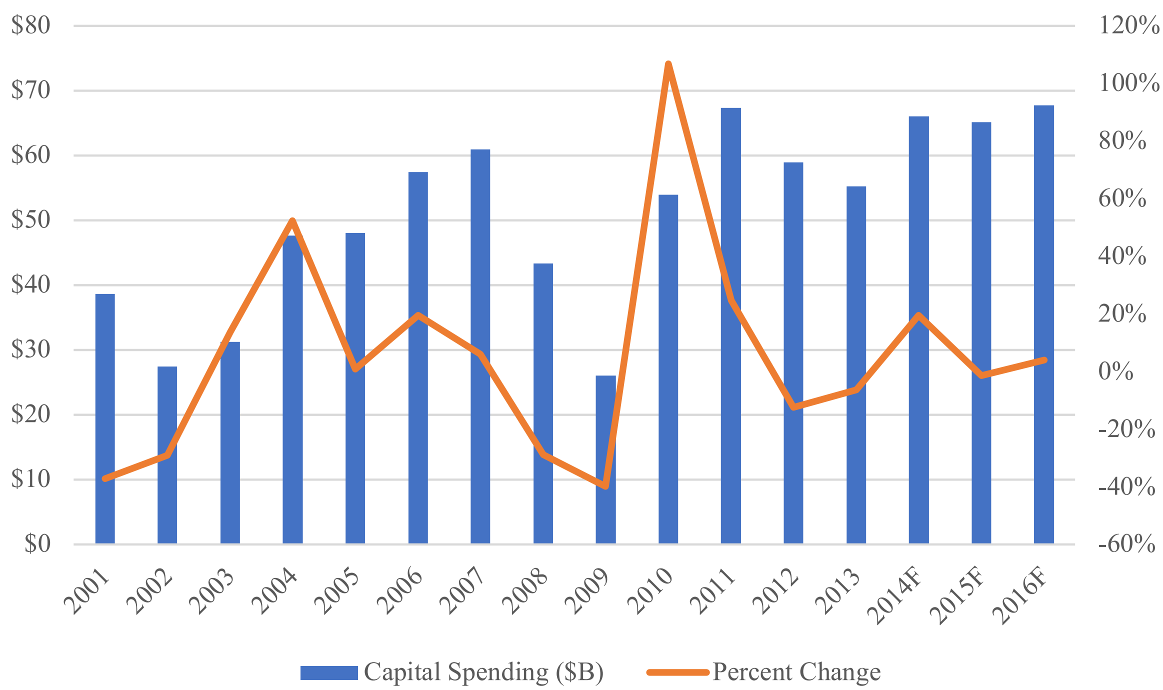
* An integrated circuit—or, for the purposes of this case, a semiconductor.

2 KLA-Tencor Annual Report 2014, https://d1io3yog0oux5.cloudfront.net/_d09e4af778fc9dc6814c73736e27f601/klatencor/db/1157/ 9978/annual_report/2014_AR.pdf.

dominating lithography, Lam and Tokyo Electron dominating the etch-and-clean segment, and KLAC dominating process control.[3]

Semiconductor capital equipment was also highly cyclical, with contractions occurring every four or five years. As **Figure 2** demonstrates, semiconductor manufacturers pulled back on capital expenditures during the macroeconomic downturn of 2008–2009, creating a backlog of demand that pushed industry-wide revenues—including those at KLAC*—sky-high for the three years between January 1, 2010, and December 31, 2012, before another natural contraction in 2013. "In this industry, a 10% contraction [like the one in 2013] is just noise," said Dirk Müller, director of marketing at Coherent, Inc., a supplier of lasers to various semicap manufacturers, including KLAC. "The semiconductor makers sometimes go on these massive buying sprees, sending the semicap company revenues through the ceiling, and then there's a natural pullback. There's always cyclicality."[4]

Figure 2: Global Semiconductor Capital Spending Trends, 2001–2016F



Source: Graph created by the case authors based on data from The McClean Report: A Complete Analysis and Forecast of the Integrated Circuit Industry.

3 Joshua Hall, "Overview of the Semiconductor Capital Equipment Industry," Seeking Alpha, January 15, 2019, https://seekingalpha.com/article/4233606-overview-of-semiconductor-capital-equipment-industry.

* KLAC's fiscal year ends on June 30, so its 2010 fiscal year includes six months from 2009 and six months from 2010. The industry figures in Figures 1 and 2 are calendar-year numbers.

4 Dirk Müller, director of marketing at Coherent, Inc., in an interview with the case authors, August 14, 2023.

KLA-Tencor

Ken Levy and Bob Anderson founded KLA Instruments Corp. in 1975 in Milpitas, California, to address the burgeoning demand for measurement and inspection tools in semiconductor manufacturing. The company first leveraged novel image processing technologies to improve yields in the chip manufacturing process; it did so through groundbreaking innovations that reduced the percentage of defective circuits. These innovations included the stylus profiler, which enabled precise surface measurement. Meanwhile, Tencor Instruments, founded in 1976 by Karel Urbanek in nearby Mountain View, California, focused on inspection and metrology systems for semiconductor wafers. Both companies went public on the NASDAQ—KLA Instruments in 1980 and Tencor in 1993—before merging in 1997 in a stock deal valued at about $1.27 billion.[5]

The merger created a powerhouse in process control solutions for semiconductor manufacturing by exploiting the synergies between KLA Instruments's proficiency in defect inspection and Tencor's expertise in surface profiling. The combined entity could now offer integrated solutions that addressed critical manufacturing challenges, positioning it as an industry leader.

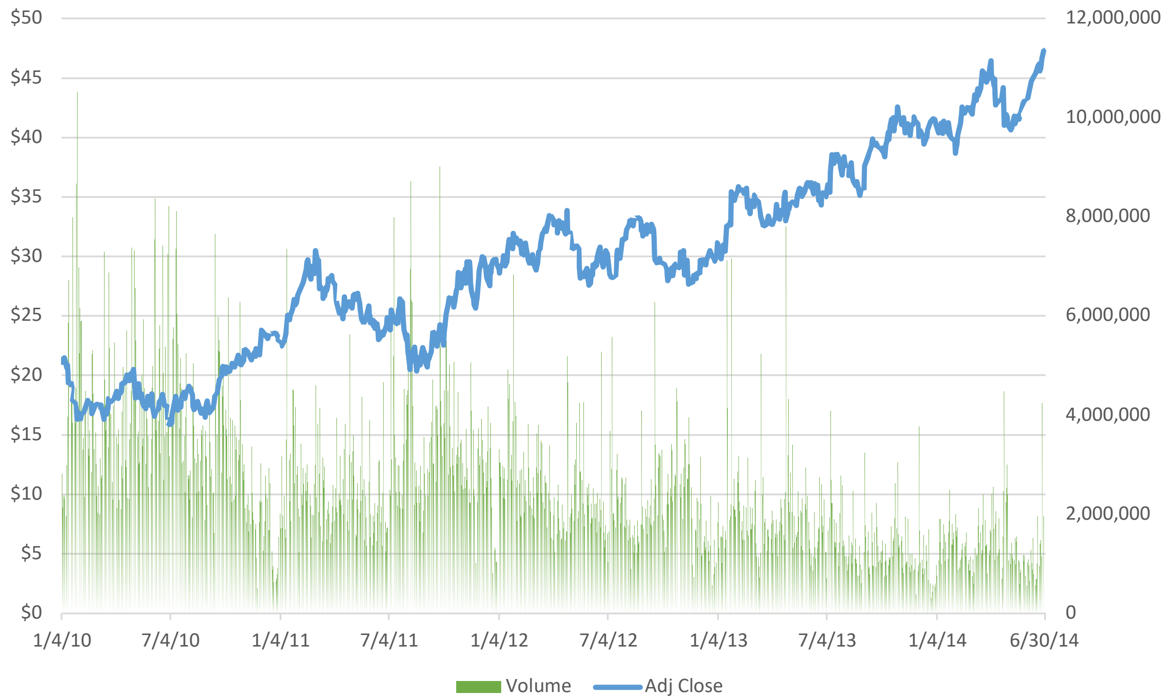
Throughout the early 2000s, KLAC expanded the breadth of its products and services, partly through a series of acquisitions. (See **Exhibit 1**.) This expansion allowed it to build a dominant brand in the process control space, with a reputation for innovation and industry leadership. "I deal with a lot of very bright management teams," Müller said. "But I think KLAC might have the best management I've worked with, the one I respect the most. Given the performance of the company's stock price, it seems like the market agrees with me."[6] (See **Figure 3**.)

This reputation for excellence drove KLAC's ongoing top-line growth, nearly doubling product revenues from $1.3 billion in 2010 to $2.3 billion in 2014 and more than doubling net income from $244 million to $617 million over the same timeframe, yielding CAGRs of 15% and 26%, respectively. (See **Exhibits 2** and **3**.) Meanwhile, cash flow from operating activities grew at a CAGR of 15% to $732 million in 2014. The backlog of demand from the 2008–2009 downturn made revenue growth choppy following the industry's cyclicality, but the company's revenue growth trendline outpaced that of the broader market, as superior execution allowed the company to steal market share from competitors.

5 "KLA Instruments, Tencor to Merge," *Los Angeles Times*, January 15, 1997, https://www.latimes.com/archives/la-xpm-1997-01-15-fi-18741-story.html; Jon Swartz, "Tencor, KLA Join Forces," *SFGate*, January 15, 1997, https://www.sfgate.com/business/article/Tencor-KLA-Join-Forces-2858549.php.

6 Müller, interview.

Figure 3: KLAC Stock Performance, January 2010–June 2014



Note: Adjusted close is the closing price after adjustments for stock splits and dividend distributions. These adjustments make the series represent share-value fluctuations better than does the closing price itself. The split ratio determines split multipliers. For example, in a 2-for-1 split, the pre-split data is multiplied by 0.5. Dividend multipliers are calculated based on dividend as a percentage of the price. For example, if a $1.25 cash dividend is distributed with an ex-dividend date of January 25, and the January 24 closing price is $51.20, then the pre-dividend data is multiplied by (1 – 1.25/51.20) = 0.9756. See https://help.yahoo.com/kb/SLN28256.html.

Source: Yahoo Finance, accessed August 14, 2023, https://finance.yahoo.com/quote/KLAC/history?p=KLAC.

The Leveraged Recapitalization

As product revenues recovered after the pullback from artificially inflated 2011 heights, market observers began to note that KLAC's stock price felt undervalued. The company performed at or near the top of its competitors—Applied Materials, Inc. (AMAT), ASML Holding NV (ASML), and Lam Research Corporation (LRCX)—in terms of EBITDA margin, operating margin, net margin, return on assets, and return on equity, and yet its price-to-earnings ratio lagged all but LRCX. (See **Exhibit 4**.)

This apparent discrepancy between price and value attracted the attention of activist hedge funds like Fletcher Capital Management. Champeau noted that each of KLAC's competitors employed significantly more leverage in their capital structures, having taken advantage of stable and historically low interest rates following the 2008–2009 macroeconomic recession. This was a recent development in the semicap equipment industry. Previously, lenders had balked at lending to an industry with moderate-to-high binary event risks and high customer concentration issues; losing just one design win with a semiconductor fabricator, for example, could sink a semicap equipment manufacturer's ability to repay debt. That changed when the Great Recession of 2008–2009 depressed stock prices throughout the semicap industry, spurring a wave of consolidation that gave players like KLAC a newfound scale and perception of safety among lenders. By 2014, lenders, struggling to find attractive yields in a low-interest-rate environment, began to see the semicap space as increasingly stable; consolidation and maturity had made the industry respectable. (See **Exhibit 5**.) While binary event risk still loomed as a potential downside catalyst to any individual issuer, industry maturation and each company's larger scale following consolidation had largely mitigated that risk in the minds of lenders, as demonstrated by the rise in issuance of "straight" (i.e., non-convertible) debt by semicap companies since 2009.

As a result, many of KLAC's competitors had taken on significant leverage over the past five years to fund either share repurchases or special (non-recurring) dividends—or both. The investment hypothesis for a repurchase typically revolved around using the debt-issuance proceeds to repurchase shares, thereby decreasing the number of shares outstanding. Such a transaction could increase the share value if the tax benefits from the new debt exceeded the costs related to the increased riskiness of the firm. While this also typically decreased the total market value of the common equity outstanding due to increased debt and the expected potential costs of financial distress—mitigated in part by the value of the tax shield—a well-executed recapitalization (or "leveraged recap" in investment banker parlance) should in theory increase the company's share price.

It was that opportunity that drove Champeau's analysis. Given the current market for debt issuance, he felt that KLAC could access debt on the scale he recommended and at the interest rates that would make the transaction make sense only by issuing out of its US entity. This means that the debt would be dollar-denominated and that Champeau would have to calculate the value of the tax shield based on the prevailing US corporate tax rate—39.1% in 2014—as opposed to some of the lower-tax overseas jurisdictions that historically had reduced its effective tax rate.

Reviewing his model's assumptions once again, Champeau stress-tested them to ensure he could defend them when presenting to management. He forecasted 11% revenue growth in fiscal year 2015, assuming the sector's cyclical recovery would continue after the 2011–2012 pullback. While this revenue growth was historically accretive to profitability metrics like EBITDA, EBIT, operating cash flow, and net income margins, Champeau took a conservative stance that the company's margins would remain flat in 2015 despite this increased contribution margin from revenue growth. He assumed that KLAC's average effective tax rate stayed the same and that operating expenses, depreciation, interest income, and working capital scaled with revenue.

Although the inconsiderable amount of long-term debt that KLAC did have (about $150 million) carried with it an interest rate of nearly 7.2%, the yield curve had both flattened and shifted such that public debt markets would be far more indulgent if Champeau could convince KLAC management to take the leap. Any interest rate would be aggressively negotiated with lenders, but Champeau's analysis of existing market comps suggested a circular relationship wherein the size of the debt would determine KLAC's ability to repay it, which would impact S&P or Moody's rating for the issuance. That rating would in turn frame the interest rate negotiation with lenders, with the agreed-upon interest rate likely determining KLAC management's appetite for issuance.

Champeau looked at four different scenarios, attempting to determine which might unlock the most shareholder value: maintaining the de minimis current level of long-term debt on the balance sheet or issuing 1.4x, 2.2x, or 3.0x the company's 2014 earnings before interest and taxes (EBIT), with all proceeds going to share repurchases, less a 0.33% placement fee for the investment bank that would run the deal.

The analysis was non-trivial. First, Champeau would have to build out a pro forma income statement for KLAC for 2015, with a placeholder for interest expense, as each scenario might produce different credit ratings for which lenders would demand different interest rates. (See **Exhibit 6**.) He would then make his best guess as to what debt rating each scenario might receive from S&P or Moody's by comparing each scenario's pro forma financials to market comps using standard metrics popular among lenders. (See **Exhibit 7**.) From the innumerable potential metrics, Champeau chose to start with three ratios: EBITDA interest coverage, operating cash flow to long-term debt, and long-term debt to assets. He forecasted operating cash flow using the indirect method by adding back depreciation and amortization to net income and subtracting the increase in net working capital.

KLAC was already rated AAA based on its de minimis long-term debt balance and strong cash flows, but Champeau recognized the possibility of that changing with a heavier debt burden. He lacked access to the algorithmic inputs that Moody's and S&P used to assign ratings to companies—and moreover, those ratings agencies tended to use overlapping ranges for metrics rather than hard cutoffs—but his close observation of the semicap equipment industry had given him sufficient confidence to estimate the approximate thresholds for each metric, as depicted in **Figure 4**.

Figure 4: Bond Ratings and Fletcher Capital Estimated Market Comp Metrics

	2015 Pro Forma						
	Credit Characteristics						
Bond Rating	**AAA**	**AA**	**A**	**BBB**	**BB**	**B**	**CCC/CC/C**
Promised Yield	4.25%	4.66%	5.47%	7.14%	9.94%	12.35%	20.44%
Expected Yield	4.21%	4.62%	5.41%	7.02%	9.53%	10.54%	10.45%
Probability of Default	0.10%	0.12%	0.16%	0.32%	1.05%	4.61%	23.70%
Market Comp Metrics							
EBITDA Interest Coverage	21.0x	16.8x	10.2x	4.9x	3.0x	1.8x	1.0x
Operating Cash Flow / Total LT Debt	152%	93%	51%	30%	19%	10%	2%
LT Debt to Asset	7.0%	11.7%	17.5%	33.8%	47.1%	69.2%	92.0%

Note: The estimates represent Champeau's prediction of the thresholds that ratings agencies use to assign a rating to a typical new debt issuance. For example, he estimated that an Operating Cash Flow / Total Long-Term Debt ratio over 152% would likely lead a ratings agency to rate an issuer as AAA, whereas he estimated that a Long-Term Debt to Asset ratio between17.5% and 33.8% would likely lead the ratings agency to rate the issuer as BBB.

Source: Default rates and yields from "Default, Transition, and Recovery: 2014 Annual Global Corporate Default Study and Rating Transitions," Standard & Poor's RatingsDirect, April 30, 2015, https://www.maalot.co.il/publications/TS20150514153957.pdf; "Doubly Bound: The Cost of Credit Ratings," Othering & Belonging Institute, accessed August 30, 2023, https://belonging.berkeley.edu/doubly-bound-cost-credit-ratings-4; and "Moody's Seasoned Aaa Corporate Bond Yield," FRED Economic Data, St. Louis Fed, accessed August 30, 2023, https://fred.stlouisfed.org/series/AAA. Ratings thresholds were estimated by the case authors for pedagogical purposes.

Next, Champeau would have to estimate the incremental present value of the tax shield in each scenario. This tax shield would increase the aggregate value available to shareholders, while his estimate of the potential costs of financial distress (CFD) would decrease it.

He grimaced at the thought. Champeau preferred clean, quantitative calculations to drive his analyses, but the potential costs of financial distress notoriously required subjective estimates. In the past, he had at times triangulated on that estimate by attempting to predict the discount that a company might have to accept on a fire sale of either inventory or fixed assets to cure a breached debt covenant, the increase in legal and advisory fees it might incur as it attempted to avoid insolvency, or the severance packages it might have to offer if it had to lay off large numbers of employees due to that distress. Champeau scratched his head over which of these, or other potential sources of financial distress, were most relevant for KLAC. Additionally, he wondered whether the CFD should vary with the size of the debt issuance.

Finally, Champeau would have to calculate the number of shares that KLAC could repurchase based on the debt proceeds (net of the placement agent's fee) and calculate the new price per share by dividing the new equity value by the reduced outstanding share count.

The Challenge

Fatigue set in; the more Champeau tried to tease the model, the harder the math became. How much value might KLAC management unlock by taking on debt to finance a share repurchase program? What rating would that debt receive from S&P or Moody's? How much leverage could the company take on without risking insolvency while maintaining its current dividend payout ratio? And how could he convince KLAC management of the viability of his plan? KLAC had already returned cash to shareholders in recent years through both dividends and some small share repurchases, but Champeau's plan would require a far more ambitious buyback plan. How could he convince management that their fiduciary duty to maximize shareholder value demanded that they take on more risk instead of continuing to play it safe?

Exhibit 1: Excerpted KLAC Acquisition History

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Year	Target	Location	Description	Consideration	Price ($MM)
1999	ACME Systems	Taiwan	Yield analysis software	Cash	6.9
2000	Finle Technologies, Inc.	Austin, TX	Lithography modeling and analysis software	Cash	5.0
2000	Fab Solutions, Inc.	Austin, TX	Advanced process control software developer	Cash	8.0
2001	Phase Metrics, Inc.	San Diego, CA	Yield management and process control	Cash and debt assumption	19.3
2004	Candela Instruments, Inc.	Manhasset, NY	Laser-based surface inspection systems	Cash	Undisclosed
2004	Wafer Inspection Systems, subsidiary of Inspex, Inc.	Bridgewater, NJ	Asset purchase of patents, software, and inventory	Cash	Undisclosed
2006	ADE Corporation	Westwood, MA	Silicon wafer metrology supplier	Cash and debt assumption	474.0
2007	OnWafer Technologies	Santa Clara, CA	Lithography and plasma etch products manufacturer	Cash	53.0
2007	SensArray Corporation	Santa Clara, CA	Temperature monitoring systems maker	Undisclosed	Undisclosed
2007	Therma-Wave Corporation	Fremont, CA	Process control and metrology manufacturer	Cash	75.0
2008	ICOS Vision Systems Corporation NV	Hong Kong	Test and measurement systems	Cash and stock	488.8
2008	Microelectronic Inspection Equipment, subsidiary of Vistec Semiconductor Systems, Inc.	Germany	Mask registration measurement and scanning electron microscopy tool provider	Cash and debt assumption	141.1
2010	Ambios Technology, Inc.	Santa Cruz, CA	Surface analysis instrument manufacturer	Undisclosed	Undisclosed
2014	Luminescent Technologies, Inc.	Palo Alto, CA	Computational lithography and inspection maker	Cash	18.0

Source: Table created by the case authors based on data from Crunchbase and Mergent Online.

Exhibit 2: KLAC Income Statement, 2010–2014

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	Year Ended June 30,				
In Thousands (USD $)	**2010**	**2011**	**2012**	**2013**	**2014**
Revenues:					
Product	$1,324,270	$2,613,438	$2,597,755	$2,247,147	$2,286,437
Service	496,490	561,729	574,189	595,634	642,971
Total revenues	1,820,760	3,175,167	3,171,944	2,842,781	2,929,408
Costs and operating expenses:					
Costs of revenues	815,662	1,259,243	1,330,016	1,237,452	1,232,962
Engineering, research and development	329,560	386,163	452,937	487,832	539,469
Selling, general and administrative	361,372	369,431	372,666	387,812	384,907
Total costs and operating expenses	1,506,594	2,014,837	2,155,619	2,113,096	2,157,338
Income from operations	314,166	1,160,330	1,016,325	729,685	772,070
Interest income and other, net	31,532	4,064	11,966	15,112	16,203
Interest expense	10,655	10,650	10,656	10,683	10,643
Income before income taxes	335,043	1,153,744	1,017,635	734,114	777,630
Provision for income taxes	90,763	327,995	227,827	156,759	160,623
Net income	$244,280	$825,749	$789,808	$577,355	$617,007

Source: KLAC 10-K Filings. Note that certain figures have been altered or simplified for pedagogical purposes.

Exhibit 3: KLAC Balance Sheet, 2010–2014

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	As of June 30,				
In Thousands (USD $)	**2010**	**2011**	**2012**	**2013**	**2014**
Current assets:					
Cash and cash equivalents	371,304	506,432	556,591	723,192	536,376
Accounts receivable, net	440,125	583,270	701,280	524,610	492,863
Inventories	401,730	575,730	650,802	634,448	656,457
Deferred income taxes	328,522	331,397	184,670	198,525	215,676
Other current assets	131,044	147,078	92,847	75,039	69,197
Total current assets	1,672,725	2,143,907	2,186,190	2,155,814	1,970,569
Land, property and equipment, net	236,752	257,358	277,686	305,281	330,263
Goodwill	328,006	328,156	327,716	326,635	335,355
Purchased intangibles, net	117,336	85,902	55,636	34,515	27,697
Other non-current assets	389,497	328,095	275,227	269,423	258,519
Total assets	2,744,316	3,143,418	3,122,455	3,091,668	2,922,403
Current liabilities:					
Accounts payable	107,938	142,945	139,183	115,680	103,422
Deferred system profit	204,764	192,338	147,218	157,965	147,923
Unearned revenue	37,026	44,264	63,095	60,838	59,176
Other current liabilities	422,059	499,314	513,411	527,049	585,090
Total current liabilities	771,787	878,861	862,907	861,532	895,611
Non-current liabilities:					
Long-term debt	145,747	146,290	146,833	147,376	147,919
Pension liabilities				57,959	59,908
Income taxes payable	53,492	78,337	50,839	59,494	59,575
Unearned revenue	20,354	34,905	34,899	42,228	57,500
Other non-current liabilities	69,065	76,235	89,235	36,616	48,805
Total liabilities	1,060,445	1,214,628	1,184,713	1,205,205	1,269,318
Stockholders' equity:					
Common stock	168	167	167	165	165
Capital in excess of par value	921,292	1,010,492	1,089,313	1,159,400	1,220,339
Retained earnings	793,714	920,530	869,405	763,544	462,852
Accumulated other comprehensive income (loss)	(31,303)	(2,399)	(21,143)	(36,646)	(30,271)
Total stockholders' equity	1,683,871	1,928,790	1,937,742	1,886,463	1,653,085
Total liabilities and stockholders' equity	2,744,316	3,143,418	3,122,455	3,091,668	2,922,403

Source: KLAC 10-K Filings. Note that certain figures have been altered or simplified for pedagogical purposes.

Exhibit 4: Comparative Data on Selected Publicly Held Semiconductor Capital Equipment Process Control Companies

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	Fiscal year ended in 2014			
	KLA-Tencor (KLAC)	Applied Materials, Inc. (AMAT)	ASML Holding NV (ASML)	Lam Research Corp (LRCX)
Summary of Operations				
Total Revenue	2,929,408	9,072,000	7,118,333	4,607,309
EBITDA	855,142	1,918,000	1,866,572	901,615
Operating Income	772,070	1,520,000	1,558,563	677,669
Interest Expense	10,643	95,000	33,615	61,692
Earnings Before Tax	777,630	1,448,000	1,548,110	723,363
Net Income	617,007	1,072,000	1,454,522	632,289
Cash Flow from Operations	732,587	1,800,000	1,246,143	717,049
Primary earnings per share	3.73	0.87	3.90	3.62
Dividends per share	1.81	0.40	0.84	0.72
Financial Position				
Cash and equivalents	536,376	3,002,000	3,213,805	3,065,644
Current Assets	1,970,569	6,967,000	8,686,248	4,783,662
Total Assets	2,922,403	13,174,000	14,833,954	7,993,306
Current Liabilities	895,611	2,823,000	3,511,437	1,582,001
Long-term Debt	147,919	1,947,000	1,397,680	817,202
Total Liabilities	1,269,318	5,306,000	5,702,365	2,963,571
Book Value of Equity	1,653,085	7,868,000	9,131,590	5,029,735
Market Value of Equity	12,017,198	27,894,460	47,113,499	12,631,718
Stock Price, June 30, 2014	72.64	22.55	93.27	67.58
Price earnings ratio	19.5x	26.0x	23.9x	18.7x
Equity Beta	1.20	1.03	1.07	0.73
Selected Ratios				
EBITDA Margin %	29.2	21.1	26.2	19.6
Operating Margin %	26.4	16.8	21.9	14.7
Net Profit Margin %	21.1	11.8	20.4	13.7
ROA % (Net)	21.1	8.1	9.8	7.9
ROE % (Net)	37.3	13.6	15.9	12.6
Operating Cash Flow / Total LT Debt	495.3%	92.4%	89.2%	87.7%
LT Debt to Book Value of Equity	8.9%	24.7%	15.3%	16.2%
EBITDA Interest Coverage	80.3x	20.2x	55.5x	14.6x
Debt Rating	AAA	A-	AA	A

Source: Table created by the case authors based on data from SEC Filings, Moody's & Fitch ratings services, and Macroaxis (beta estimates). Note that certain figures have been altered or simplified for pedagogical purposes.

Exhibit 5: Semiconductors as % of World GDP, 1984–2022F

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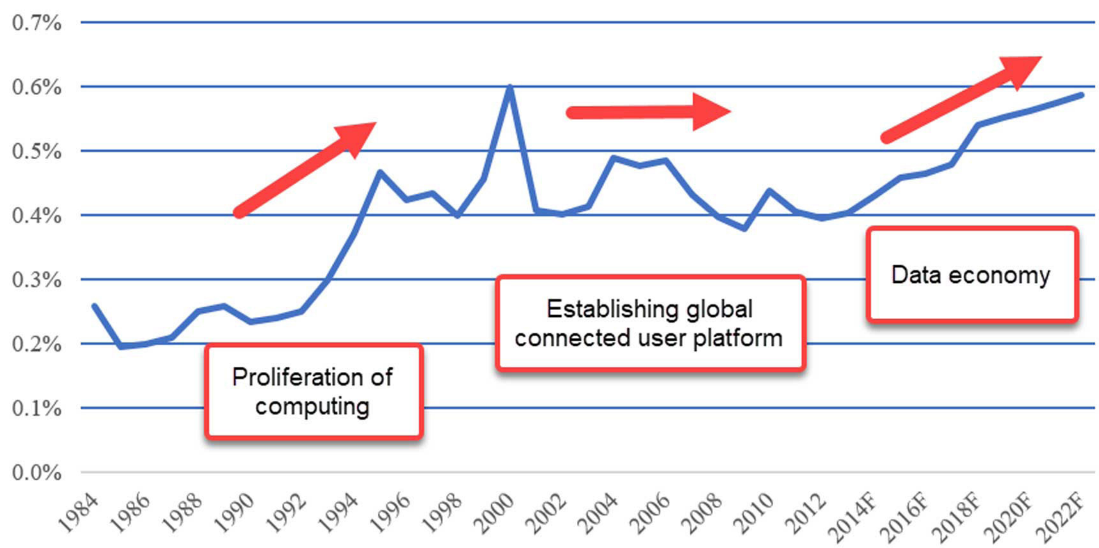

Source: Graph created by the case authors based on a report by Lam Research.

Exhibit 6: Pro Forma Results for Alternative Capital Structures

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		2015 Pro Forma			
	2014	**Current Capital Structure**	**Scenario I**	**Scenario II**	**Scenario III**
Debt Issuance (multiple of EBIT)		0.0x	1.4x	2.2x	3.0x
Debt Issuance ($)		$0	$1,080,898	$1,698,554	$2,316,210
Revenue growth		11%	11%	11%	11%
Total revenue	2,929,408				
Cost of revenue	1,232,962				
Gross profit	1,696,446				
Operating expenses	841,304				
EBITDA	855,142				
Depreciation and amortization	83,072				
EBIT	772,070				
Interest income	16,203				
Interest expense	10,643				
Profits before taxes	777,630				
Taxes	160,623				
Profits after taxes	617,007				
Depreciation and amortization	83,072				
Increase in net working capital	(32,508)				
Operating Cash Flow	732,587				
Total LT Debt	147,919				
Market Value of Assets	13,286,516				
Book Value of Assets	2,922,403				
Market Value of Equity	12,017,198				
Book Value of Equity	1,653,085				
Shares Outstanding	165,435				
Share Price	72.64				

Source: Table created by the case authors.

Exhibit 7: Standard & Poor's Debt Ratings Definitions and Answers to FAQs

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Here is the general meaning of our credit rating opinion:

- 'AAA'—Extremely strong capacity to meet financial commitments. Highest rating.
- 'AA'—Very strong capacity to meet financial commitments.
- 'A'—Strong capacity to meet financial commitments, but somewhat susceptible to adverse economic conditions and changes in circumstances.
- 'BBB'—Adequate capacity to meet financial commitments, but more subject to adverse economic conditions.
- 'BBB-'—Considered lowest investment grade by market participants.
- 'BB+'—Considered highest speculative grade by market participants.
- 'BB'—Less vulnerable in the near-term but faces major ongoing uncertainties due to adverse business, financial, and economic conditions.
- 'B'—More vulnerable to adverse business, financial, and economic conditions but currently has the capacity to meet financial commitments.
- 'CCC'—Currently vulnerable and dependent on favorable business, financial, and economic conditions to meet financial commitments.
- 'CC'—Highly vulnerable; default has not yet occurred but is expected to be a virtual certainty.
- 'C'—Currently highly vulnerable to non-payment, and ultimate recovery is expected to be lower than that of higher-rated obligations.
- 'D'—Payment default on a financial commitment or breach of an imputed promise; also used when a bankruptcy petition has been filed or similar action taken.

Note: Ratings from 'AA' to 'CCC' may be modified by the addition of a plus (+) or minus (-) sign to show relative standing within the major rating categories.

Are Credit Ratings Indicators of Investment Merit?

While investors may factor credit ratings into their investment decisions, Standard & Poor's ratings are *not* indications of investment merit. In other words, ratings are not buy, sell, or hold recommendations or a measure of asset value. And they are not intended to signal the suitability of an investment for any person or institution. They speak to one aspect of an investment decision—credit quality—which, in some cases, may include our view of what investors can expect to recover in the event of default.

Exhibit 7 *(continued)*

Credit ratings are just one of many factors—including the current make-up of their portfolios, their investment strategy and time horizon, their tolerance for risk, and an estimation of the securities' relative value in comparison to other securities they might choose—that we expect market participants to consider when evaluating debt securities. By way of analogy, while reputation for dependability may be an important consideration in buying a car, it is not the sole determinant in the purchase decision. It depends on individual needs and requirements.

Are Credit Ratings Absolute Measures of Default Probability?

Because no one can predict the future, the assignment of credit ratings is not an exact science. For this reason, Standard & Poor's ratings opinions are not intended as guarantees of credit quality or as exact measures of the probability that a particular issuer or particular debt issue will default. Instead, ratings express relative opinions about the creditworthiness of an issuer or credit quality of an individual debt issue, from strongest to weakest, within a universe of credit risk. The likelihood of default is the single most important factor in our assessment of creditworthiness.

For example, a corporate bond that is rated 'AA' is viewed by Standard & Poor's as having a higher credit quality than a corporate bond with a 'BBB' rating. But the 'AA' rating isn't a guarantee that it will not default, only that, in our opinion, it is less likely to default than the 'BBB' bond.

Why Do Credit Ratings Change?

"The only thing constant is change"—and credit ratings can change.

Reasons for ratings adjustments vary and may be broadly related to overall shifts in the economy or business environment or more narrowly focused on circumstances affecting a specific industry, entity, or individual debt issue.

In some cases, changes in the business climate can affect the credit risk of a wide array of issuers and securities. For instance, new competition or technology, beyond what might have been expected and factored into the ratings, may hurt a company's expected earnings performance, which could lead to one or more rating downgrades over time. Growing or shrinking debt burdens, hefty capital spending requirements, and regulatory changes may also trigger ratings changes.

Although some risk factors may affect all issuers—an example would be growing inflation that affects interest rate levels and the cost of capital—other risk factors may pertain only to a narrow group of issuers and debt issues. For instance, the creditworthiness of a state or municipality may be impacted by population shifts or lower incomes of taxpayers, which reduce tax receipts and ability to repay debt.

Source: "Ratings Definition FAQs," Standard & Poor's Ratings Services, accessed September 12, 2023, https://web.archive.org/web/20140703030539/https://www.spratings.com/about/about-credit-ratings/228950121.html.